

Zhichao, Huang

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EDUCATION

East China Normal University, Shanghai, China

09/2024 - 07/2027 (expected)

- School of Computer Science and Technology

• M.Eng. in Computer Technology

GPA: 3.66/4.0

- Advisors: Prof. Gao Yang (<https://ygao36buffalo.github.io/>); Prof. Li Yang (<https://ihpdep.github.io/>)

Southwest University, Chongqing, China

09/2020 - 06/2024

- College of Artificial Intelligence

• B.Eng. in Intelligent Science and Technology

CGPA: 3.54/4.0

RESEARCH INTERESTS

• Human-computer interaction, Wearable device design, Multi-sensor interaction

Current research focuses on a strain-gauge force-sensor-based force-feedback master-slave robotic arm system.

PUBLICATIONS

- **ZhiChao Huang***, Yingnian Guo*, Chiyue Wang, Zhengte Cai, Xiongfeng Ying, Yu He, Yingjing Xiao, KaiyiGuo, Qian Zhang, Zhanpeng Jin, and Yang Gao. EmbodiedRecall: A Ring-to-Glasses System for Preserving Valuable, Fleeting Moments in Daily Activities. accepted pending minor revisions. (*Equal contribution)
- Yingjing Xiao*, **Zhichao Huang***, Junbin Ren, Yuting Bai, Haichuan Song, Zhanpeng Jin, and Yang Gao. 2025. Wrist2Finger: Sensing Fingertip Force for Force-Aware Hand Interaction with a Ring-Watch Wearable. In Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST '25). Association for Computing Machinery, New York, NY, USA, Article 24, 1–16. <https://doi.org/10.1145/3746059.3747767>. (*Equal contribution)
- Yingjing Xiao*, **Zhichao Huang***, and Yang Gao. 2025. From Wrist to Finger: Hand Pose Tracking Using Ring-Watch Wearables. In Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25). Association for Computing Machinery, New York, NY, USA, Article 294, 1–7. <https://doi.org/10.1145/3706599.3720220>. (*Equal contribution)

RESEARCH EXPERIENCE

• Vision-Based Tactile Sensor Integration for Robotic Grasping

07/2026

Advisor: Professor Li Yang, East China Normal University, Shanghai, China

---- Integrated vision-based tactile sensors into robotic grippers and designed sensor mounting structures for the end effector of an ALOHA robotic arm;

---- Assembled a dual-gripper tactile sensing prototype with one vision-based tactile sensor mounted on each side; completed structural calibration, leveling, and gripper motion-space validation;

---- Built a tactile image acquisition and visualization workflow during grasping, validating data stability and contact-state representation through object-grasping experiments.

• Robot-Mediated Interaction and Force-Feedback VR Research

05/2026 - 07/2026

Advisor: Professor Li Yang, East China Normal University, Shanghai, China

- Designed a robotic gripper force-feedback system for human-robot interaction, integrating piezoresistive force sensors with mechanical mounting, embedded sensing/control, and force-feedback algorithms;
- Participated in force-sensor signal-chain design and validation, focusing on the complete path from sensing element, signal conditioning, and ADC acquisition to MCU/host-side processing;
- Designed and iterated the gripper-side sensor mounting structure and VR-controller-side force-feedback mechanism using SolidWorks; completed prototype assembly, leveling, and system integration;
- Built the force-feedback mapping pipeline between the controller and robotic gripper, combining sensor data, servo/actuator control, and wireless communication for real-time grasp feedback;
- Completed the controller-robotic-arm teleoperation pipeline and continued improving the force-feedback mechanism; completed a paper submitted to UbiComp;
- Optimized mechanical structure, sensor placement, and feedback-control strategy to improve feedback stability, response speed, and interaction naturalness.
- **Linux Edge Voice Interaction Device Development and Deployment** 07/2025 - 03/2026
Project with Prof. Shaohui Lin's EIC Lab: Edge Intelligent Computing Lab
 - Evaluated the Rockchip RK3588 Linux platform and ReSpeaker Mic Array v2.0 audio front end for edge voice interaction device development and deployment;
 - Built a local speech recognition and speaker verification prototype, including audio front-end integration, on-device model deployment, voice-command recognition, and response feedback;
 - Integrated a 10.1-inch touchscreen and validated touchscreen driver, click input, and embedded interaction interface functionality.
- **Smart Glasses Prototype Triggered by Ring Gesture Recognition** 09/2025 - 07/2025
Advisor: Professor Gao Yang, South China University of Technology, Shanghai, China
 - Developed ESP32-S3 smart glasses prototypes with OV2640/OV5640 cameras, Wi-Fi image transmission, microSD storage, and video clip recording;
 - Built a ring-glasses interaction pipeline, enabling on-device IMU inference of the ring and control of the glasses for photo and video capture;
 - Reduced the wearable camera module to a 35 mm × 10 mm form factor; assembled a complete battery-powered system weighing 6.2 g with a 150 mAh battery;
 - Integrated large language model-assisted capture-timing prediction to support context-aware user recording reminders.
- **Wireless Watch-Mounted EMG Sensing Module for Hand Interaction** 05/2025 - 01/2025
Advisor: Professor Gao Yang, South China University of Technology, Shanghai, China
 - Designed and validated a piezoresistive fingertip pressure sensing system for five-finger grasping tasks, supporting grasp-force estimation and force-aware hand interaction modeling;
 - Developed a NRF52832-based wireless EMG module integrating an INA333AIDGKR instrumentation amplifier, LMV358IDT dual operational amplifiers, and gold-plated dry electrodes for weak EMG acquisition;
 - Participated in multi-channel sensor data acquisition, collecting EMG, finger IMU, and fingertip-force data with synchronized recording, preprocessing, and visualization;
 - Completed a UIST 2025 paper using IMU and EMG signals to predict fingertip force during grasping.
- **Bluetooth IMU Ring Prototypes for Wearable Hand Interaction** 09/2024 - 11/2024
Advisor: Professor Gao Yang, South China University of Technology, Shanghai, China
 - Designed multiple compact Bluetooth IMU rings using NRF52832, ESP32-S3, and EFR32BG22C224F512 platforms, integrating ICM20948 IMU sensing, battery-powered embedded circuits, and 3D-printed or FPC-based enclosures;

- Implemented low-power wearable hardware iterations from a 9.5 mm-wide NRF52832 prototype to an 8 mm-wide modular EFR32/FPC design, reducing average operating current to 8-9 mA;
- Integrated Wi-Fi/Bluetooth communication, a button, and a micro vibration motor into the ESP32-S3 version for gesture-triggered reminders, user confirmation, and wireless data interaction;
- Completed a CHI EA 2025 paper using ring IMU data for hand pose tracking, and completed a paper submitted to CHI on gesture-triggered capture with ring-glasses interaction.

PROFESSIONAL EXPERIENCE

- Research Assistant, Chongqing Lingyou Technology Co., LTD.** 09/2020 - 09/2024
Advisor: Prof. Songhua Yang, College of Artificial Intelligence, Southwest University, Chongqing, China
 ---- Participated in engineering projects, gaining experience in embedded system design, hardware prototyping, server maintenance, and project collaboration;
 ---- Designed capacitive touchscreen circuits, Altium Designer PCB schematics/layouts, and SolidWorks mechanical enclosures for laser rangefinder products; participated in motor-control hardware integration;
 ---- Participated in embedded product solution development and hardware debugging, gaining practical experience in sensors, motor control, touch interaction, and product prototype integration.
- FPGA Engineering Intern, Intel China Innovation Center** 06/2023 - 11/2023
 ---- Participated in the FPGA-based implementation of a LeNet-5 neural-network accelerator, completing hardware acceleration workflow development, functional validation, and performance optimization for handwritten digit recognition and face-mask detection tasks.

AWARDS & HONORS

- First-Class Academic Scholarship for Graduate Students, East China Normal University, Academic Year 2024–2025.
- Undergraduate Scholarship, Southwest University, Academic Year 2022–2023.
- Undergraduate Scholarship, Southwest University, Academic Year 2020–2021.

Technical Skills

- **Embedded and Hardware Circuits:** Circuit design, PCB/FPC design, sensor integration, and wireless communication; familiar with ESP32 series and EFR32BG22 Bluetooth chips; experience with NRF52832 Bluetooth chips, STM32 series chips, and Rockchip RK3588.
- **Sensors and Signal Processing:** IMU sensors, EMG sensors, piezoresistive/strain-gauge force sensors, and vision-based tactile sensors; experience with sensor signal acquisition chains, instrumentation amplifiers, ADC acquisition, filtering, calibration, and data visualization.
- **Robotics and Communication Systems:** Experience with robotic grippers, force-feedback control, wireless data communication, and host/edge-device system integration; understanding of low-latency and reliable sensor data links.
- **Linux Edge Device Development:** Experience with Linux edge-device prototyping, local model deployment, audio front-end integration, Socket/network communication, and touchscreen-based embedded interaction.
- **Mechanical Design:** Familiar with SolidWorks; experience in designing and assembling wearable device enclosures, robotic gripper sensor mounts, and force-feedback interaction mechanisms.